WEEKLY ASSIGNMENT

Python – Strings | Answers with explanations and programs

1. What is a string? How can strings be created in Python?

A string is a sequence of characters enclosed in quotation marks. Strings can be created using single quotes, double quotes, or triple quotes.

```
name = 'Python'
message = "Hello World"
paragraph = '''This is
a multi-line string.'''
```

2. What is string indexing? Explain positive and negative indexing.

String indexing is used to access an individual character in a string. Positive indexing starts from 0 at the beginning. Negative indexing starts from -1 at the end.

```
word = "PYTHON"
print(word[0])    # P (positive index)
print(word[2])    # T
print(word[-1])   # N (negative index)
print(word[-2])   # O
```

3. What is string slicing?

String slicing is used to extract a part of a string. Its syntax is string[start:stop:step]. The start index is included, but the stop index is excluded.

```
word = "PYTHON"
print(word[1:4])   # YTH
print(word[:3])    # PYT
print(word[2:])    # THON
print(word[::-1])  # NOHTYP
```

4. Write a program to find the length of a user-entered sentence.

The len() function returns the total number of characters in a string, including spaces and punctuation.

```
sentence = input("Enter a sentence: ")
print("Length of the sentence:", len(sentence))
```

5. Write a program to count the number of vowels in a string.

The program checks each character and increases the count if it is a vowel. lower() makes the check case-insensitive.

```
text = input("Enter a string: ")
count = 0

for ch in text.lower():
    if ch in "aeiou":
        count += 1

print("Number of vowels:", count)
```

6. Write a program to reverse a string.

String slicing with [::-1] returns the characters in reverse order.

```
text = input("Enter a string: ")
reversed_text = text[::-1]
print("Reversed string:", reversed_text)
```

7. Write a program to check whether a word is a palindrome.

A palindrome is a word that reads the same forward and backward, such as 'level'. The program compares the word with its reverse.

```
word = input("Enter a word: ")

if word == word[::-1]:
    print("It is a palindrome")
else:
    print("It is not a palindrome")
```

8. Write a program to count how many times a particular character appears in a string.

The count() string method returns the number of occurrences of a specified character or substring.

```
text = input("Enter a string: ")
ch = input("Enter the character to count: ")

print("Occurrences:", text.count(ch))
```

9. Explain the difference between upper, lower, title and capitalize string operations.

upper() converts all letters to uppercase. lower() converts all letters to lowercase. title() converts the first character of each word to uppercase. capitalize() converts only the first character of the string to uppercase and the remaining letters to lowercase.

```
text = "welcome to PYTHON programming"

print(text.upper())       # WELCOME TO PYTHON PROGRAMMING
print(text.lower())       # welcome to python programming
print(text.title())       # Welcome To Python Programming
print(text.capitalize())  # Welcome to python programming
```

10. Write a program to remove extra spaces from the beginning and end of a sentence.

The strip() method removes whitespace from the beginning and end of a string. It does not remove extra spaces between words.

```
sentence = input("Enter a sentence: ")
cleaned = sentence.strip()
print("After removing leading and trailing spaces:", cleaned)
```